AF-3849 **MPH1C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Physics

MATHEMATICAL PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Divergence of a vector function.

2. State stoke's theorem.

3. Define Vector spaces.

4. Examine the linear independence of vectors

$\overrightarrow{x_1} = (1,0,0)$, $\overrightarrow{x_2} = (0,1,0)$ and $\overrightarrow{x_3} = (0,0,1)$.

5. What are Eigenvalues and Eigenvectors ?

6. List any four properties of Rank of Matrix.

7. Write the Sturn-Liouville Differential equation and its boundary Conditions.

8. State Cauchy's residue theorem.

9. List the rules of Singular points.

10. Define Fourier Sine series.

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Show that div grad $\phi = \Delta^2\phi$.

(*Or*)

(b) Verify Stoke's theorem for the vector

$A = (2x - y)\hat{i} - yz^2\,\hat{j} - y^2 z\,\hat{k}$ over the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$.

12. (a) State and prove Schwartz inequality.

(*Or*)

(b) Find the Eigenvalues and Eigenvectors of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

13. (a) Solve $x^2 \, {d^2 y}/{dx^2} - x\dfrac{dy}{dx} + y = 2\log x$.

(*Or*)

(b) Show that the equation ${dy}/{dx} + Py = Qy^n$ is reduced to a linear equation by the substitution $\upsilon = y^{1-n}$.

14. (a) State and prove Cauchy's Integral formula.

(*Or*)

(b) State and prove Laurent's theorem.

15. (a) Find the Complex form of the Fourier series of $f(x) = e^{-x}$ in $-1 \le x \le 1$.

(*Or*)

(b) Develop the Fourier Series for

$$f(x) \begin{cases} 0 & \text{when} \quad -\pi < x \le 0 \\ x & \text{when} \quad 0 \le x < \pi \end{cases}.$$

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove Gauss divergence theorem.

17. Diagonalise the following Matrix :

$$\begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

18. Obtain the solution of linear Second Order Differential Equation with Constant and Variable Coefficients.

19. Evaluate : $\int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4 \cos \theta} \, d\,\theta.$

20. Obtain the Fourier series for $f(x) = x^2$ in the interval $-\pi < x < \pi$.

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AF-3850 **MPH1C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Physics

CLASSICAL DYNAMICS AND RELATIVITY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. State D' Alembert's Principle.

2. What are generalised coordinates ?

3. State Kepler's third law of Planetary motion.

4. Define Differential scattering Cross - section.

5. Define Euler's angles.

6. What is meant by normal mode of Vibration ?

7. State the principle of Least action.

8. Define Poisson bracket.

9. State the two basic postulates of special theory of relativity.

10. What are the conditions for the Lorentz transformation ?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Develop the notation for the displacement in terms of generalised coordinates.

(*Or*)

(b) Derive the Lagrange's equation of motion from Hamilton's principle for a conservative system.

12. (a) Explain briefly about the two main features of the motion of a particle under the action of a central force.

(*Or*)

(b) Deduce an expression for the orbit under inverse square law of force.

13. (a) What are moments of inertia coefficients and products of inertia ?

(*Or*)

(b) Write a short note on Wave Packet.

14. (a) Give the equations of motion in Poisson bracket form.

(*Or*)

(b) Explain briefly about the Action angle variable and mention one use of this formalism.

15. (a) Discuss the Relativistic law of Addition of velocities.

(*Or*)

(b) Derive Einstein's Mass–energy relation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What are Constraints ? Explain how they can be classified.

17. State and prove the Virial theorem.

18. What are normal Coordinates ? Discuss the normal frequencies of a vibrating motion for a linear triatomic molecule.

19. Give the solution of a harmonic oscillator problem by Hamilton - Jacobi method.

20. (a) Discuss briefly about the Kinematic effects of Lorentz transformation.

(b) Prove that the four dimensional volume element '$dx\ dy\ dz\ dt$' is invariant under Lorentz transformation.

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AF-3851 **MPH1C3**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Physics

ELECTRONICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Mention some applications of the Schottky diode.

2. What is a photodiode ? Explain.

3. Explain the basic construction of a N-chemical JFET.

4. Explain two transistor analogy of a SCR.

5. Draw the circuit symbol and pin configuration of a Op-Amp.

6. What is a voltage follower ? Describe the main characteristics.

7. Explain the term resolution for DAC.

8. Mention the difference between triangular and sawtooth waves.

9. What is an integrated circuit ? Explain the different scale of integration.

10. Mention some applications of CCD's.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Briefly explain the behaviour of junction in forward and reverse bias mode.

(*Or*)

(b) With a neat sketch, explain the working of a LASER diode.

12. (a) Draw and explain the V-I characteristics of UJT. Mention some applications.

(*Or*)

(b) Explain the construction and working of Triac.

13. (a) Elaborate the various Op-Amp parameters.

(*Or*)

(b) Draw and explain the circuit diagram of an integrator using OP-Amp.

14. (a) Explain the following :

(a) Schmitt's trigger.

(*Or*)

(b) Counter-ramp ADC.

15. (a) List out the advantages and disadvantages of an IC's.

(*Or*)

(b) Explain the working of IC 555 timer as Astable muitivibrator.`

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss the tunneling phenomenon with the help of energy and band diagrams. Briefly explain the construction and working of a Tunel Diode.

17. Briefly discuss the working, V-I characteristics and applications of JFET.

18. With the help of circuit diagram and gain versus frequency graph, explain the following filters :

    (a) Low pass filter.

    (b) High pass filter.

19. Draw the circuit diagram and explain the action of phase-shift oscillator.

20. Describe the different steps involved in fabricating diodes, transistors and inductors.

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AF-3852 **MPH1E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Physics

Elective—NUMERICAL METHODS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Absolute Error.

2. What is the principle on which the method of group averages is based ?

3. The order of convergence of Newton-Raphson method is ————.

4. What is iteration method ?

5. State Lagrange's interpolation formula.

6. Write down formula for $(y')_{x = x_n}$ using Newton's backward difference formula.

7. State Simpson's $1/3$ and $3/8$ rule.

8. Give the order of the error in Simpson's rule.

9. Write down the algorithm of Improved Euler's method.

10. Using Runge-Kutta method of second order, find $y\,(0.1)$ given $y' = -y,\ y\,(0) = 1$.

Part B

$(5 \times 5 = 25)$

Answer **all** questions

11. (a) By the method of group averages, fit a straight line to the following data :

x :	0	5	10	15	20	25
y :	12	15	17	22	24	30

(*Or*)

(b) Derive the normal equations to fit straight line.

12. (a) Find the root of the equation $x^2 - 3x = 2$ in the vicinity of $x = 0$ using Newton-Raphson method.

(*Or*)

(b) Solve the following system of equation by Gauss-Seidel method, correct to three decimal places.

$$x + y + 54z = 110$$

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

13. (a) Obtain an expression for Newton's forward interpolation formula.

(*Or*)

(b) Apply Gauss's forward central difference formula and estimate $f(32)$ from the following table.

x	:	25	30	35	40
$y = f(x)$	:	0.2707	0.3027	0.3386	0.3794

14. (a) From the following table of values of x and y, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for $x = 1.05$

x :	1	1.05	1.1	1.15	1.2	1.25	3
y :	1.0000	1.0247	1.04881	1.07238	1.09544	1.11803	1.14017

(*Or*)

(b) Write the program in C language for evaluating the Integral using Simpson's $1/3$ rule.

15. (a) Using Improved Euler's method, find the value of y when $x = 0$ given that $y(0) = 1$ and $y' = x^2 + y$.

(*Or*)

(b) Write the C program for solving first order differential equation by Runge-Kutta method.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Fit a parabola, by the method of least squares to the following data, also estimate y at $x = 6$.

x	:	1	2	3	4	5
y	:	5	12	26	60	97

17. Write the C program for finding roots using Newton-Raphson method.

18. Find the value of $\cos 51^\circ 42'$ by using Gauss's backward interpolation formula from the table given below :

x	:	50°	51°	52°	53°	54°
$y = \cos x$	:	0.6428	0.6293	0.6157	0.6018	0.5878

19. Find the numerical value of d^2y/dx^2 at $x = 0.4$ of the function defined as under :

x	:	0.1	0.2	0.3	0.4
$f(x)$	:	1.10517	1.2214	1.34986	1.49182

20. Find y (0.1), y (0.2) given $y' = x - 2y$, y (0) = 1 taking $h = 0.1$ by third order Runge-Kutta method.

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AF-3853 **MPH1E2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

First Semester

Physics

Elective—CRYSTAL GROWTH AND THIN FILMS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the advantages of solution growth ?

2. What do you mean by seasoning ?

3. What is the structure of gel ?

4. Mention any two advantages of gel growth.

5. Mention the importance of crucible shape.

6. Which factors affect the growth from melt ?

7. Define Sputter yield.

8. What is the principle of flash evaporation ?

9. Mention the advantages of XRD.

10. What is etching ?

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Explain seed preparation.

(*Or*)

(b) Describe Slow cooling method.

12. (a) List down the importance of gel.

(*Or*)

(b) Write a note on complex-decomplexion method.

13. (a) Briefly explain Vertical Bridgmann technique.

(*Or*)

(b) Write a note on Transport material and Transport agents.

14. (a) Discuss about eraporation sources.

(*Or*)

(b) Write a note on Spray Pyrolysis.

15. (a) Explain the principle and working of Vickers micro hardening tester.

(*Or*)

(b) Briefly explain the working of SEM.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Explain the working of constant temperature bath.

(b) Write a note on crystallizer.

17. Write notes on

(a) Double diffusion method.

(b) Chemical reduction method.

18. Discuss the chemical vapour deposition method in crystal growth and mention its advantages and limitations.

19. Write notes on radio frequency sputtering and resistive heating.

20. Explain the principle and working of FTIR Technique.

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AF-3854 **MPH2C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Physics

ELECTROMAGNETIC THEORY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State and explain Coulomb's law.

2. Derive Poisson's equation.

3. Define Ampere's law in Circuital Form.

4. Write a note on Magnetic Permeability.

5. Define Poynting Vector.

6. State Conservation of energy in electromagnetism.

7. Explain Penetration depth.

8. Define Anisotropic and Isotropic medium.

9. Define Differential scattering cross - section.

10. Why does the sky appear blue ? Explain.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Derive differential form of Gauss's law.

(*Or*)

(b) Show that $\text{div } J + \frac{\partial e}{\partial t} = 0.$

12. (a) Prove that at the end of the magnetic field in a long solenoid is half of that in the middle of the solenoid.

(*Or*)

(b) Write short note on Magnetic induction and Magnetic susceptibility.

13. (a) Derive the relation between the Electric field E and the Changing magnetic field B with the help of Faraday's law.

(*Or*)

(b) Derive the intergral form of Maxwell's equation and explain their physical significance.

14. Write a short note on the following :

(a) Total internal reflection.

(*Or*)

(b) Group velocity.

15. Briefly explain the following :

(a) Electric and Magnetic Dipole radiation.

(*Or*)

(b) Multipole expansion for localised source.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Write down Laplace's equation in spherical co–ordinates and obtain its solution. What will be the form of solution for spherically symmetric problems ?

17. Differentiate between Magnetic scalar and Vector potentials. Derive an expression for the magnetic vector potential due to a small element of a current carrying circuit at a large distance.

18. What are gauge transformations ? Derive and discuss about Coulomb's gauge.

19. Discuss the reflection and refraction of electromagnetic waves at the interface of two non – conducting media and derive Fresnel equations.

20. Briefly discuss the Rayleigh's scattering of radiation by electrons. Why does the rising or setting sun appear red ?

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AF-3855 **MPH2C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Physics

QUANTUM MECHANICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What do you mean by the expectation value of a dynamical quantity ?

2. State Ernfest theorem.

3. What is the significance of zero point energy ?

4. State the eigen functions for the states Ψ_{200} and Ψ_{211} in hydrogen atom.

5. What is a harmonic perturbation ?

6. Define Transition probability.

7. Find the value of $[J_z, J_+]$.

8. Give the eigenvalues of J^2 and J_2.

9. What is Lamp shift ?

10. Give the value of D' Alembertian operator.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Derive time dependent Schrödinger wave equation.

(*Or*)

(b) Write a note on Hilbert space.

12. (a) Solve the Schrödinger wave equation for a free particle in three dimension.

(*Or*)

(b) Obtain energy eigenvalue expression of one dimensional harmonic oscillator.

13. (a) Write a note on adiabatic approximation.

(*Or*)

(b) Calculate the first order Stark effect for the non excited state ($n = 2$) in hydrogen atom.

14. (a) Explain the properties of a C.G. coefficient.

(*Or*)

(b) Discuss the scattering of identical particles.

15. (a) Write a note on negative energy states.

(*Or*)

(b) Write a note on spring orbit coupling.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Solve Schrödinger wave equation for a particle in square well potential and find eigenvalue and eigenfunctions.

17. Solve the Schrödinger wave equation for hydrogen atom and explain the state of degeneracy.

18. Discuss the first order time dependent perturbation theory and derive Fermi's Golden Rule.

19. Give the theory of Born approximation in scattering calculations. Using it find the scattering cross section for an electron nuclei.

20. Obtain the free particle solutions of Dirac's relativistic equation and discuss the negative energy state.

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AF-3856 **MPH2C3**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Physics

SOLID STATE PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is called the bars ?

2. Define the Symmetry operations.

3. What is a lattice plane ?

4. Define the elastic constants for a crystal.

5. What is an Intrinsic Semiconductor ?

6. Define Carrier concentration.

7. What do you mean by the Polarization of a Solid ?

8. What are the characteristics of ferromagnets ?

9. Write short notes on Persistent currents.

10. What is the role of electrical resistivity in super conductors ?

Part B $(5 \times 5 = 25)$

Answer **all** the questions

11. (a) Give the summary of seven crystal systems and fourteen Bravais Lattices

(*Or*)

(b) Write a note on Bravais Lattice.

12. (a) Describe the characteristics of strain-stress relations in cubic crystals.

(*Or*)

(b) Show that the frequencies for modes of propagation vectors K and K + G are the same where G is a reciprocal lattice vector.

13. (a) Derive law of mass action for semi conductors ? What is its practical utility ?

(*Or*)

(b) Discuss the role of impurities in electrical conductivity.

14. (a) Discuss the Macroscopic electrical field.

(*Or*)

(b) Explain the Rare earth ions in the Dielectrics.

15. (a) Describe the effect of magnetic field in a superconductor.

(*Or*)

(b) Describe the microwave and infrared properties of superconductors.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What are the colour centers ? How are they produced ? Explain.

17. Describe the inelastic Scattering of neutrons for the experimental determination of phonon spectra.

18. Write briefly the Tight bound approximation in band theory of solids.

19. Explain the difference between :

(a) Diamagnetism and Paramagnetism.

(b) Paramagnetism and Ferromagnetism.

(c) Ferromagnetism and Antiferromagnetism.

20. Derive London equations and account for high frequency effects on superconductors.

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AF-3857 **MPH2E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Physics

Elective—MATHEMATICAL METHODS IN PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. List any two properties of Fourier transform.

2. Define Laplace transform.

3. Write the Wave equation and Diffusion equation in partial differential form.

4. Give the Laplace equation in different Co-ordinate systems.

5. Define Rank of a Tensor.

6. Show that the law of transformation for a Contravariant vector is transitive.

7. Define permutation groups.

8. What are the Homomorphism and Isomorphism ?

9. Define Beta function.

10. State Orthogonality theorem.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) If $f(x) = \begin{cases} x & \text{for } 0 \le x \le \frac{1}{2}\pi \\ \pi - x & \text{for } \frac{1}{2}\pi \le x \le \pi, \end{cases}$

express it by a sine series.

(*Or*)

(b) Find the inverse Laplace transform of :

(i) $\dfrac{1}{(s+1)(s+2)^2}$.

(ii) $\dfrac{s+2}{(s^2+4s+5)^2}$.

12. (a) $\dfrac{d^2y}{dx^2} - y = x \sin x + (1+x^2)e^x$.

(*Or*)

(b) Obtain the solution for one dimensional heat equation.

13. (a) Prove that Kronecker delta is a mixed tensor of rank two.

(*Or*)

(b) Explain what is meant by Covariant, Contravariant and mixed tensors. If $\beta_{\upsilon\sigma}$ is an arbitrary Covenant tensor and $A(\mu, \upsilon)\beta_{\gamma\sigma} \subset C_{\dot{\mu}\sigma}$ where $C_{\mu\sigma}$ is a tensor, then prove that $A(\mu, \upsilon)$ is a mixed Tensor.

14. (a) State the basic postulates of group. Show that the order of every sub group of a main group is a divisor or order of group.

(*Or*)

(b) Derive formula to show that the number of time an Irreducible representation would occur in the regular representation.

15. (a) Prove that :

(i) $Q_n(1) = \alpha$.

(ii) $Q_n(\alpha) = 0$.

(*Or*)

(b) Derive the relation between Beta and Gamma function.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Obtain the general solution for the transverse vibrations of stretched string.

17. Derive an expression for temperature distribution of the plate in the transient state.

18. (a) Explain Metric tensor.

(b) Surface of a sphere is a two dimensional Riemann space. Find its fundamental metric tensor.

19. Obtain Irreducible representation of a point group C_{3V} and construct its character table.

20. Obtain the solution for Bessel's Differential equation.

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AF-3858 **MPH2E2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Second Semester

Physics

Elective : NANO TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Why metals are good conductors ?

2. Give two examples each for top - down and bottom - up approaches.

3. What is Plasma ?

4. Define Masking.

5. What are the different types of nanotubes ?

6. What are lipids ?

7. Define Self assembly.

8. What are optoelectronic devices ?

9. State Moore's law.

10. What are the advantages of MEMS ?

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Explain the significance of arrangements of elements in the periodic table.

(*Or*)

(b) Explain how the chemical properties of materials vary with surface to bulk ratio.

12. (a) What are aerogels ? Mention their applications.

(*Or*)

(b) Describe nanomaterial fabrication by plasma arcing.

13. (a) Discuss how carbon nanotubes are separated from other entities.

(*Or*)

(b) Explain the function of DNA as wire template.

14. (a) What are self assembled monolayers ? Explain their applications.

(*Or*)

(b) What are applications of nanotechnology in imaging technology ?

15. (a) Explain electron beam lithography method for nanofabrication.

(*Or*)

(b) Explain how is a Quantum computer different to a Classical computer.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay on different types of nanomaterials, their properties and potential applications.

17. Describe Electrodeposition and Ball milling methods for nanomaterial fabrication.

18. Explain the production of nanotubes by Laser methods and Chemical vapour deposition.

19. Explain the interaction of light at the nanoscale and applications.

20. What are quantum computers ? Explain their experimental implementations.

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AF-3859 **MPH3C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

STATISTICAL PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give the concept of entropy.

2. State the first law of thermodynamics.

3. What is mean free path ?

4. What is Collision probability ?

5. Define Micro state and Macro state.

6. Write down the expression for Maxwell-Boltzmann distribution law of energies.

7. What is fermi energy ?

8. What is meant by exchange degeneracy ?

9. Write down the drawbacks of Einstein's theory of specific heat.

10. State any two properties of Fermi ideal gas.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Derive Clausius Dapeyron latent heat equation.

(*Or*)

(b) Distinguish between Adiabatic and Isothermal process.

12. (a) Write a note on Hydrodynamics.

(*Or*)

(b) State the general features of kinetic theory.

13. (a) Write a note on micro canonical ensemble.

(*Or*)

(b) Explain phase space.

14. (a) Write a note on density matrix.

(*Or*)

(b) Briefly explain Fermi Dirac's statistics.

15. (a) Discuss about liquid helium.

(*Or*)

(b) Explain gas degeneracy.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly discuss the transport phenomena and derive an expression for viscosity of a gas.

17. Explain what do you understand by Gibb's function. Show that the change in Gibb's function is zero for reversible isothermal and isobaric processes.

18. Prove the conservation principle of density in phase space.

19. Explain the phenomenon of Bose Einstein condensation and discuss.

20. Explain the Einstein's theory of specific heat of solids.

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AF-3860 **MPH3C2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

NUCLEAR AND PARTICLE PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are magic numbers ?

2. What are nuclear forces ?

3. What do you mean by nuclear isomerism ?

4. What are particle detectors ?

5. Explain Q-value of nuclear reactions.

6. Explain Nuclear chain reaction.

7. Define Cold fusion.

8. What is meant by direct reactions ?

9. State CPT theorem.

10. What are Quarks ?

Part B (5 × 5 = 25)

Answer **all** the questions

11. (a) What are exchange forces ? Explain their types.

(*Or*)

(b) Describe Yukawa's Meson theory of Nuclear forces.

12. (a) Describe the principle, construction and working of a Scintillation counter.

(*Or*)

(b) State and explain Selection rules for-β-decay.

13. (a) Explain the principle of Cyclotron and mention its limitations.

(*Or*)

(b) Explain controlled thermonuclear reactions.

14. (a) Write a note on Compound nucleus model for Nuclear reactions.

(*Or*)

(b) Write a note on Direct reactions.

15. (a) Explain CP violation in neutral K-Meson decay.

(*Or*)

(b) Discuss various types of interactions involved in elementary particles.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Give an account of the nuclear shell model. What are its successes and limitations ?

17. Describe the construction and working of semi-conductor counter. What are the special advantages of this counter over others ?

18. Give the Bohr-Wheeler's theory of Nuclear fission.

19. Obtain Briet-Wigner's dispersion formula for resonance reaction.

20. (a) Write a detailed note on the classification of elementary particles.

(b) Discuss Su (3) classification scheme of elementary particles.

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AF-3861 **MPH3E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

Elective—ENERGY PHYSICS AND ENVIRONMENTAL SCIENCE

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are Fossil fuels ?

2. Name the places in Tamil Nadu wherein wind energy is produced.

3. Define heat convection.

4. What is stellar energy ?

5. What are High temperature collectors ?

6. Point out the basic elements of a solar water heater.

7. Give the principle of SPV.

8. Define Efficiency of Solar Cell.

9. What are contaminants ?

10. Does UV radiation deplete Ozone layer ? Explain.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) What are fuel cells ? Explain.

(*Or*)

(b) Write a note on Biomass.

12. (a) Describe the working of a Pyroheliometer.

(*Or*)

(b) How do you convert solar radiation into heat ?

13. (a) What are concentrators ? Explain.

(*Or*)

(b) How does a Solar water heater differ from an Electrical heater ?

14. (a) Give the V-I characteristics of a Solar cell.

(*Or*)

(b) What are the uses of Solar cells ?

15. (a) What are the sources of air pollution ?

(*Or*)

(b) What are the impacts of water pollution ?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the functioning of OTEC plant.

17. Explain the principle and working of flat plate solar collectros.

18. Write notes on :

(a) Concentrating collector.

(b) Evacuvated tube collector.

19. What are SPV cells ? Explain their functioning.

20. Describe the role of purification and control devices in water and air pollution.

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AF-3862 **MPH3E2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

Elective : COMMUNICATION ELECTRONICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Frequency Modulation.

2. Write the power gain of Antenna.

3. Define Pulse Code Modulation.

4. What is time division multiplexing ?

5. What is Klystron ? List the different Klystrons.

6. Give radar equation.

7. Write different types of fiber.

8. What are fiber joints ?

9. How the satellite communication frequencies are assingned ?

10. Write a difference between Analog and Digital cellular mobile systems.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write the theory of Phase Modulation.

(*Or*)

(b) Briefly explain vertical and loop ferrite rod antenna.

12. (a) Write about digital carrier system.

(*Or*)

(b) Write a note on Amplitude Shift Keying.

13. (a) How the Microwaves are detected ?

(*Or*)

(b) Write about MTI radar.

14. (a) Write a short note on Optical detectors.

(*Or*)

(b) Give some Splicing techniques.

15. (a) Write briefly about Satellite construction.

(*Or*)

(b) Explain Multiple Access Cellular Systems.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss :

(a) Antennas and its equivalent circuits.

(b) Power gain of Antenna.

(c) UHF and VHF antennas.

17. Write in detail

(a) Error control coding.

(b) Multiplex transmission.

(c) Delta modulation.

(d) Pluse frequency modulation.

18. Explain how the Reflex and Multicavity Klystron are working with necessary diagrams.

19. How does the signal degradation of fibers occur ? Also discuss scattering losses.

20. Give the general principle for cellular system and write its operation and planning with a neat diagram.

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AF-3863 **MPH3E3**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

Elective—MICROPROCESSORS AND MICROCONTROLLERS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Mention the names of various registers in 8085 along with its size.

2. What is the function of program counter ?

3. What are the different types of data transfer scheme ?

4. What are the operating modes of Port-A of 8255 PPI ?

5. List the number of register banks and their starting addresses.

6. Which bits in which register must be set to give the serial data interrupt the highest priority ?

7. What are the contents of A register upon RESET of 8051 μC ?

8. Which register bank shares the same space with the stack ?

9. Which pin of the ADC 804 indicates end-of-conversion ?

10. Give the 4-step sequence of a stepper motor if we start with 0 1 1 0.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the functions of ALE and $IO/\overline{M}$ signals of 8085 μP.

(*Or*)

(b) Compare CALL-Return and PUSH-POP instructions of 8085 μP.

12. (a) Explain Interrupt driven data transfer scheme using flow chart.

(*Or*)

(b) Draw the internal block diagram of 8255 PPI.

13. (a) Write a note on Special Function Registers of 8051 μC.

(*Or*)

(b) Explain the Mode 1 operation of timer register of 8051 μC.

14. (a) Explain the difference between carry and overflow flags of 8051 μC with examples.

(*Or*)

(b) Write a program to add two 8-bit numbers using 8051 μC with Register and Immediate addressing modes.

15. (a) Write a note on DAC.

(*Or*)

(b) Briefly explain how Seven Segment Display interfaced with 8051 μC.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the instruction set of 8085 μP .

17. Draw the functional block diagram of 8251 A and explain the function of each block.

18. Explain the key features and architecture of 8051 μC using suitable diagram.

19. Explain various types of addressing modes of 8051 μC with examples.

20. Write a note on the following 8051 μC applications :

    (a) Pressure measurement

    (b) Water level indicator.

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AF-3864 **MPH3E4**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Third Semester

Physics

Elective—DIGITAL SIGNAL PROCESSING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give the applications of DSP.

2. What are the various types of representations of discrete time signal ?

3. Define Z-transform.

4. Define Stability.

5. List the applications of DFT.

6. List the properties of linear convolution.

7. Mention any two properties of Chebyshev filter.

8. List the features of II R filters.

9. Give two examples of fixed point DSP.

10. List the functions of shifter.

Part B (5 × 5 = 25)

Answer **all** questions

11. (a) Explain the classification of discrete time signals.

(*Or*)

(b) Explain any *two* operations on discrete time signals.

12. (a) Briefly explain system function.

(*Or*)

(b) Write a note on Linearity and Time sharing.

13. (a) State and prove shifting property of DFT.

(*Or*)

(b) What is zero padding ? Why it is needed ?

14. (a) Write a note on magnitude and phase response of digital filters.

(*Or*)

(b) Explain the function of Butterworth filters.

15. (a) Write a note on COMBOCODEC Interface with DSP.

(*Or*)

(b) Write a note on Radar signal processing using DSP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Draw and explain the block diagram of DSP and also mention some of the advantages and disadvantages of DSP.

17. Explain various properties of Z-transform.

18. Discuss the properties of FFT.

19. Explain :

(a) Inverse Chebyshev filters.

(b) Elliptic filters.

20. Explain the architecture of DSP using suitable block diagram and also list the salient features.

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AF-3865 **MPH4C1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Physics

ATOMIC AND MOLECULAR PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Give the principle of Stren and Gerlach experiment.

2. What do you mean by spin-orbit interaction ?

3. What is Zeeman effect ?

4. What are molecular orbitals ?

5. Are the rigid rotator energy levels degenerate ? Explain.

6. Explain why IR frequencies are generally expressed in terms of wave numbers.

7. How does Raman effect differ from fluorescence ?

8. What are selection rules ?

9. What is PMR ?

10. Define the term "g" factor.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe and explain L–S coupling.

(*Or*)

(b) State and explain Pauli's exclusion principle. How does a knowledge of symmetric and antisymmetric wave functions lead to this principle ?

12. (a) State and explain Stark effect.

(*Or*)

(b) Discuss briefly the Hückel's molecular approximation.

13. (a) Describe in brief the rotational spectra of Polyatomic molecules.

(*Or*)

(b) Derive an expression for the energy of symmetric top molecules.

14. (a) Describe in brief the electronic spectra of diatomic molecules.

(*Or*)

(b) Write four applications of Raman effect explaining how it is used in the study of molecular structure.

15. (a) Write notes on :

(i) Spin-Spin.

(ii) Spin-Lattice relaxation times.

(*Or*)

(b) What is chemical shift ? Explain.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What is meant by fine structure of hydrogen spectra ? How is it explained on the basis of spin orbit interaction ?

17. Explain Heilter - London theory for the formulation of Hydrogen molecule.

18. Give the theory of diatomic Vibrating rotator.

19. State and explain Franck - Condon principle.

20. Describe the experimental set up for studying ESR.

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AF-3866 **MPH4E1**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Physics

Elective : MEDICAL PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the value of normal blood pressure ?

2. What is the effect of passing a DC current through tissue ?

3. What are hearing aids ?

4. Define Ultrasound.

5. What do you mean by bloodless surgery ?

6. For what colour has the eye the best sensitivity ?

7. How X rays are produced ?

8. What are dose units ?

9. State Bernoulli's Principle.

10. What are electrodes ?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe how Lung volumes are measured.

(*Or*)

(b) How do you realise the existence of current in our body ?

12. (a) Describe the functioning of a Stethoscope.

(*Or*)

(b) What are the physiological effects of ultrasound in therapy ?

13. (a) List the applications of IR in the field of medicine.

(*Or*)

(b) Write a note on optical illusions.

14. (a) How do you record X ray image ?

(*Or*)

(b) What are the side effects of radiotherapy ?

15. (a) Explain how we measure direct cardiac pressure using a Catheter.

(*Or*)

(b) Describe the functioning of Defibrillators.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What is an Electro Cardiogram ? Draw a typical ECG waveform, label all waves, and explain. What is happening within the heart muscle during each wave?

17. Describe Anatomy and Physiology of the Ear.

18. Describe the principle and sample preparation for a SEM.

19. How accurately does a radiotheraphy dose have to be specified ? What criteria would you see to select suitable isotope sources for radiotherapy and for radionuclide imaging ?

20. Describe the functioning of a Pacemaker.

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AF-3867 **MPH4E2**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Physics

Elective—BIOMEDICAL INSTRUMENTATION

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Resting Potential.

2. What are surface electrodes ?

3. What is Phonocardiograph ?

4. Define Latency.

5. What are the different types of heart sounds ?

6. What is Cardiac output ?

7. What is Larmor frequency ?

8. What are the different magnets that can be used in NMP imaging system ?

9. Define Pace maker.

10. Mention the different types of diathermy.

Part B (5 × 5 = 25)

Answer **all** the questions questions.

11. (a) Discuss the different sources of biomedical signals.

(*Or*)

(b) Explain EMG electrodes in detail.

12. (a) Explain in detail Vector Cardiograph.

(*Or*)

(b) Explain the description of an Electro Oculagraphy.

13. (a) Explain the measurement of blood Pressure by direct method.

(*Or*)

(b) Explain the working of NMR blood flowmeter.

14. (a) Explain the different detection coils used in NMR imaging system.

(*Or*)

(b) Discuss the different arrangements in multi-element linear array Scanners.

15. (a) Explain the working of Synchronised d.c. fibrillator with a neat block diagram.

(*Or*)

(b) Explain with Schematic diagram the working of microwave diathermy unit.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write short notes on :

(a) Nervous system.

(b) Silver-Silver chloride electrodes.

17. Explain the working of ECG instrumentation with block diagram.

18. Explain the working of laser doppler blood flow meter with a neat block diagram.

19. Describe the Ultrasonic imaging System (M-mode) with a suitable diagram.

20. Discuss the various types of implantable pacemakers.

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